Automated Camera Inspection System

Comprehensive Technical Documentation

End-of-Line Automotive Seat Quality Inspection

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1.1 What is the System?

The **Automated Camera Inspection System** is an end-of-line (EOL) quality inspection solution for automotive seat manufacturing. It performs automated visual inspection at four sequential stations along the production line, using a combination of machine vision algorithms and deep-learning AI models to detect cosmetic defects, verify component presence, and validate mechanism operation — all without slowing production throughput.

Each seat travels through four stations. Results from all stations are aggregated by the **Inspection Coordinator**, which drives a safety interlock: the OEM label printer is energised *only* when every station confirms a clean pass, the recliner mechanism cycle passes, and the lock sensor is confirmed. Any failure holds the pallet and diverts it to the reject spur.

1.2 Key Capabilities

Capability	Description
Multi-Station Pipeline	Four independent inspection stations covering leather surface, stitching, plastics, and full mechanism cycle.
AI-Powered Detection	ONNX inference with PatchCore anomaly detection (Stn 1) and YOLO defect detectors (Stn 2-4).
Shadow Mode	Run the full pipeline in observation-only mode — no physical PLC outputs — to validate AI before go-live.
Human Review Queue	Borderline detections enter a review queue; an inspector confirms PASS / REWORK / SCRAP.
Immutable Audit Trail	Every decision, reviewer action, and system event is timestamped and stored in an append-only log.
Real-Time Dashboard	Live seat-strip, quality trends, defect gallery, shift reports, SPC charts, and system health.
Shadow Mode Analytics	Tracks AI vs inspector agreement rate, escape rate, and false-reject rate per shift.
Image Traceability	Raw and annotated images saved per seat per camera, exportable as a ZIP audit package.
Flexible Hardware	Camera adapters for Basler, Hikrobot, webcam, phone, RTSP; PLC adapters for Modbus, Snap7, OPC-UA.
Active Learning Pipeline	Inspector feedback on false positives is stored as JSONL for future model retraining.

Component	Technology
Backend API	Python 3.13+ / FastAPI / Uvicorn
Database	SQLite via SQLAlchemy ORM (PostgreSQL-ready)
AI Inference	ONNX Runtime — PatchCore + YOLO models
Classical Vision	OpenCV
PLC Communication	Simulator / Modbus / Snap7 / OPC-UA
Camera SDKs	pypylon (Basler), MVS SDK (Hikrobot), OpenCV (webcam)
Frontend	React + TypeScript + Vite + Recharts

API Documentation	Swagger UI at /docs
Config Format	YAML (system, stations, variants)

2.1 High-Level Architecture

The diagram below represents the top-level data and control flow. Physical pallets travel left-to-right through the four stations. Each station runs independently and reports to the central Coordinator.

Component	Role	Interacts With
Station 1-4	Capture, inspect, report	Camera, PLC, DB, Coordinator
Inspection Coordinator	Aggregate results, gate label printer	All stations, PLC, DB
Line Controller	Conveyor, E-Stop, diverter	Conveyor, PLC, Operator Panel
PLC Layer	Physical signal I/O	Conveyor, Printer, Sensors
Database (SQLite)	Persist all records	All services
FastAPI Server	REST API + dashboard	Coordinator, DB, Frontend
React Frontend	UI — live view, reports, settings	FastAPI

2.2 Inspection Flow

A single seat traversal follows this ordered sequence:

Step	Name	Description
1	RFID Write	Seat ID written to RFID tag on pallet.
2	Station Trigger	PLC signals each station when the pallet arrives.
3	Barcode Read	Station reads barcode to resolve seat ID and variant.
4	Variant Load	Variant config and limits loaded; LimitsEvaluator constructed.
5	Fixture Clamp	Fixture clamps pallet; clamp sensor confirmed before capture.
6	Lighting Apply	Per-camera lighting profile applied and read-back verified.
7	Image Capture	All enabled cameras capture; any failure → FAIL immediately.
8	AI Inference	ONNX model runs on each captured frame.
9	Limits Evaluate	Measurements compared against pass/review/fail limits.
10	DB Persist	Result persisted BEFORE PLC is notified (fail-safe order).
11	PLC Notify	Station outcome written to PLC tag (skipped in shadow mode).
12	Coordinator	All-station results aggregated; printer gating evaluated.
13	Route Pallet	PASS → mainline + label; FAIL/REVIEW → reject spur.

Inspection Coordinator

The **InspectionCoordinator** service (coordinator/service.py) is the safety-critical aggregation engine. Key responsibilities:

- Buffer partial station results in memory until all four stations report.
- Enforce the printer interlock: enable ONLY when all stations PASS, mechanism PASSES, and lock sensor is confirmed.
- Run a background watchdog thread — seats that do not complete within cycle_timeout_s are auto-finalised as FAIL.

• Auto-disable the printer after print_pulse_timeout_s (5 s default) to prevent spurious prints.

• In shadow mode, skip all PLC calls and queue every result for human audit.

The **LineController** (line/controller.py) manages physical material flow. It triggers E-Stop (stops conveyor, holds all pallets, kills printer, lights red indicator), orchestrates the per-seat cycle across all four stations, and routes the pallet to the mainline or reject spur based on the aggregated outcome.

Every station inherits from **BaseStation** (station/base_station.py). This abstract class handles: hardware pre-check, PLC trigger wait, barcode parsing, multi-camera capture sequence, lighting verification, calibration enforcement, fail-safe exception handling, DB persistence, and PLC notification.

3.1 What is Shadow Mode?

Shadow Mode is a production-safe validation operating mode where the full camera inspection pipeline runs — capturing images, running AI models, computing defect decisions, and storing results — but *no physical outputs are driven*. The PLC is not commanded, pallets are never held, and the label printer is never energised.

■ *Shadow mode is the recommended approach before any go-live on a new line, after a major AI model update, or when introducing a new seat variant.*

3.2 Normal Mode vs Shadow Mode

Feature / Action	Normal Mode	Shadow Mode
AI inference runs	✓ Yes	✓ Yes
Images captured	✓ Yes	✓ Yes
Database records written	✓ Yes	✓ Yes
Audit log written	✓ Yes	✓ Yes
Defects detected & stored	✓ Yes	✓ Yes
PLC station result written	✓ Yes	✗ Skipped
Pallet hold activated	✓ Yes on FAIL	✗ Never
Label printer triggered	✓ Yes on PASS	✗ Never
Conveyor controlled	✓ Yes	✗ Never
PASS seats queued for audit	On request	✓ Always
Shift report generated	✓ Yes	✓ Yes
Agreement tracking	No baseline	✓ AI vs Inspector

3.3 Behaviour In Detail

Coordinator in Shadow Mode

When shadow_mode=True is set on the **InspectionCoordinator**, the finalize logic still fully evaluates the overall outcome and printer eligibility — it simply skips the plc.set_label_printer_enable() and plc.hold_pallet() calls. Every finalised seat is automatically queued for human review so an inspector can record their own PASS/FAIL decision.

Station in Shadow Mode

Each station checks self.shadow_mode at step 11 (PLC Notify). When true, the step is entirely skipped and a log line is written: *"Shadow mode active: PLC interlock skipped for outcome FAIL"*. All other pipeline steps — capture, inference, DB write — execute normally.

```
# Relevant code in base_station.py (step 12)
if not self.shadow_mode and self.plc is not None:
self.plc.set_station_result(
self.station_id, result.outcome, seat_id=result.seat_id
)
else:
logger.info(
f'Shadow mode active: PLC interlock skipped '
f'for outcome {result.outcome}'
```

```
)
```

3.4 Configuration

```
system:
environment: production
shadow_mode: false # false = normal, true = shadow
operating_mode: shadow # 'shadow' or 'camera_leads_with_audit'
save_images: true
image_retention_days: 30
```

```
# Enable shadow mode
POST /api/v1/system/shadow-mode
Content-Type: application/json

{"enabled": true}

# Disable shadow mode
{"enabled": false}
```

The toggle propagates immediately to the coordinator and every running station service. The change is also recorded in the audit log.

```
python -m camerainspection.main \
--config configs/system.yaml \
--plc-adapter simulator
```

3.5 Shadow Mode Analytics

While shadow mode is active, human inspectors record their own judgement via the API. The system then tracks four KPIs:

KPI	Definition	Ideal Target
Agreement Rate	% of seats where AI and inspector agree	> 95%
Escape Rate	AI says PASS, inspector says FAIL (missed defect)	< 0.5%
False Reject Rate	AI says FAIL, inspector says PASS (over-rejection)	< 3%
Review Rate	Proportion sent to human review queue	< 5%

All four KPIs are visible in real time on the **Shadow Mode Statistics** dashboard view (GET /api/v1/dashboard/shadow-mode) and in the end-of-shift report.

Scenario	Recommendation
New line commissioning	Run shadow for ≥ 1 week before enabling outputs
AI model update	Shadow mode for ≥ 500 seats before switching live
New seat variant introduction	Shadow for full pilot batch to tune variant limits
Post-maintenance restart	Shadow for one shift to confirm hardware stability
OEM audit preparation	Run shadow in parallel to generate comparison data
Software update deployment	Shadow for first shift after update

4.1 Station 1 — Leather / Rexene Surface Inspection

Station 1 inspects dark leather and rexene upholstery panels for cosmetic surface defects. It uses a combination of dome and low-angle raking lights to reveal texture anomalies, paired with a PatchCore anomaly detection model trained exclusively on defect-free reference images.

Station ID	STATION_1
Purpose	Dark surface anomaly detection + segmentation
Camera Resolution	2592 × 1944 px
Pixel Size	0.18 mm/px
Exposure	15 000 µs
Lighting — Dome	80% intensity
Lighting — Low-angle	60% intensity
AI Model	Anomalib PatchCore (ONNX)
Pass Threshold	score < 0.40
Review Band	0.40 – 0.60
Fail Threshold	score ≥ 0.60
Trigger Timeout	5.0 s

front_cushion_backrest	Seating face — main inspection area for scratches, cuts, pinholes.
rear_back_map_pocket	Rear panel — side back and map pocket area.

- Scratch
- Cut / laceration
- Pinhole
- Stain / contamination
- Foam showthrough
- Wrinkle / puckering
- Colour deviation

Station 2 inspects the sewn seams across bolsters, headrest, and arm using high-resolution cameras at 0.08 mm/px — fine enough to resolve individual stitches. A YOLO model detects broken threads and skip stitches; rule-based measurement verifies stitch density (SPI) and seam position.

Station Parameters

Station ID	STATION_2
Purpose	Stitch defect detection + dimensional seam measurement
Camera Resolution	2592 × 1944 px
Pixel Size	0.08 – 0.09 mm/px (fine pitch)

Lighting	Direct 90% intensity
AI Model	YOLO stitch defect (ONNX)
Confidence Threshold	0.50
Trigger Timeout	5.0 s

left_bolster	Left bolster seam — main_seam ROI.
right_bolster	Right bolster seam — main_seam ROI.
headrest	Headrest seam — main_seam ROI.
seam_arm	Robot-arm camera with configurable poses (bolster_start / bolster_end).

Measurement	Nominal	Pass Tolerance	Fail Threshold	Unit
Stitches Per Inch (SPI)	7.0	±0.5	±1.0+	spi
Seam Position Deviation	0.0	±1.5 mm	> ±2.0 mm	mm
Stitch Pitch Delta	0.0	±0.5	> ±1.0	spi
Thread Colour Match	Variant BGR	Within tolerance	Out of range	–

4.3 Station 3 — Plastic Parts & Shield Inspection

Station 3 verifies that all plastic components (side shield, lever handle, ISOFIX covers, clips) are present, in the correct hand (LH/RH), correctly coloured, and free of cracks or short shots. A golden-image NCC comparison is used alongside the YOLO presence model.

Station ID	STATION_3
Purpose	Part presence, hand, colour, crack, short-shot detection
Camera Resolution	2592 × 1944 px
Pixel Size	0.18 – 0.22 mm/px
Lighting	Diffuse 85 – 90%
AI Model	YOLO presence (ONNX)
Confidence Threshold	0.40
Golden Match Threshold	0.85 (NCC)
Colour Check	Enabled
Trigger Timeout	5.0 s

Camera Views

side_shield_handle_lever	Side shield ROI + lever/handle ROI.
underside	Underside clips area.

Components Verified (FRONT_LH_BLACK variant)

Component	Expected Hand	Required	Count
side_shield	LH	Yes	1
handle	LH	Yes	1

lever	LH	Yes	1
isofix_cover	–	Yes	2
clips	–	Yes	6

4.4 Station 4 — Complete Seat & Mechanism Cycle Test

Station 4 performs the final full-seat cosmetic check *and* exercises the recliner and track mechanism through a full travel cycle. The lock sensor must be confirmed before the label printer can ever be enabled. This is the critical safety gate for OEM release.

Station ID	STATION_4
Purpose	Cosmetics + recliner/track mechanism cycle + lock confirmation
Camera Resolution	2592 × 1944 px
Pixel Size	0.15 mm/px
Lighting	Ambient 100%
AI Model	YOLO cosmetic defects (ONNX)
Confidence Threshold	0.45
Mechanism Full Cycle	Yes — full recliner travel commanded
Sensor Poll Timeout	10.0 s
Min Lock Torque	15.0 Nm
Max Slip Back	1.0 mm
Trigger Timeout	12.0 s

Measurement	Nominal	LSL	USL	Unit
Recliner Angle	90.0	88.0	92.0	deg
Track End Position	240.0	238.0	242.0	mm

■ **Station 4 result is required for label printer enable. A missing or FAIL result at Station 4 prevents label printing regardless of other station outcomes.**

5.1 Camera Adapters

The system supports eight camera adapters. The active adapter is set per camera in the station YAML and can be overridden at startup via --camera-adapter.

Adapter	Use Case	Notes
replay	Development / CI	Loops pre-recorded PNG/JPG from a directory.
synthetic	Unit testing	Generates solid-colour or pattern frames; no hardware needed.
webcam	Development	USB webcam via OpenCV; device index configurable.
phone	Development	Android phone via DroidCam PC client.
rtsp	Production	Any RTSP-capable network camera.
ip	Production	Direct IP camera HTTP stream.
basler	Production	Basler industrial cameras via pypylon SDK.
hikrobot	Production	Hikrobot / MindVision cameras via MVS SDK.

5.2 PLC Integration

A single PLC abstraction layer (hardware/plc/base.py) supports four physical adapters, selectable via the plc.adapter config key or the --plc-adapter CLI flag.

Adapter	Protocol	Typical Use
simulator	In-process	Development, CI, shadow validation.
modbus	Modbus TCP	Generic industrial PLCs on port 502.
snap7	S7 ISO	Siemens S7-300/400/1200/1500 series.
opcua	OPC-UA	Modern unified automation standard.

PLC Functions

- **wait_for_trigger(station_id, timeout_s)** — block until the pallet-arrival trigger fires.
- **read_barcode(station_id)** — read barcode string from the PLC register.
- **set_station_result(station_id, outcome)** — write PASS/FAIL/REVIEW tag to PLC.
- **hold_pallet(station_id, hold)** — assert/de-assert the pallet hold solenoid.
- **set_label_printer_enable(enabled)** — pulse the printer output relay.
- **is_mechanism_locked()** — read recliner lock sensor (digital input).
- **simulate_arrival(station_id, barcode)** — simulator helper to inject a trigger.
- **is_connected()** — liveness check used by system health dashboard.

Modbus-based lighting controllers are supported via hardware/lighting/modbus_lighting.py. Each camera profile specifies per-channel intensity percentages; the station verifies the read-back within ±5% before capture — failing verification raises a HardwareError and fails the seat safely.

dome	Uniform overhead dome light — reduces shadowing.
low_angle	Side / raking light — emphasises surface texture.

direct	Focused collimated beam — highlights stitch detail.
diffuse	Even diffuse panel — removes glare on plastics.
ambient	General area illumination.

5.4 RFID & Barcode

An RFID writer (hardware/rfid/) writes the seat ID to a tag on the pallet when the cycle starts. At each station a barcode scanner (or PLC register) provides the barcode string. The system resolves the **seat_id** and **variant_id** using this priority order:

- Colon separator — SEAT_ID:VARIANT_ID
- Known-variant suffix match — SEAT_ID_FRONT_LH_BLACK
- Bare variant barcode — whole barcode is a known variant ID
- Fallback underscore split — SEAT_ID_

5.5 Conveyor & Diverter

The LineController calls the conveyor adapter to physically route each pallet after the full inspection cycle completes:

release_to_mainline(seat_id)	PASS seat — released to production mainline.
divert_to_reject(seat_id)	FAIL or REVIEW seat — diverted to reject/rework spur.
stop()	E-Stop — immediate halt of all conveyor movement.

5.6 Operator Panel

An operator panel with three indicator lights provides shop-floor status at a glance.

Green (solid)	Seat PASSED — pallet released to mainline.
Red (solid)	Seat FAILED or REVIEWING — pallet held on reject spur.
Amber (solid)	System in standby or E-Stop just reset.
Red + Green off	E-Stop active — line halted.

6.1 AI Models

PatchCore is an unsupervised anomaly detection algorithm from the Anomalib library. The model is trained on defect-free reference surfaces only. At inference it returns a pixel-level anomaly map and a scalar anomaly score (0 = normal, 1 = highly anomalous). The model is exported to ONNX for cross-platform deployment.

Model file	models/station_1/patchcore.onnx
Input	2592 × 1944 RGB image
Output	anomaly_score (float), anomaly_map (heatmap)
Pass	score < 0.40
Review	0.40 ≤ score < 0.60
Fail	score ≥ 0.60
Training	camerainspection/training/train_anomalib.py

YOLOv8-based models are exported to ONNX and run via the OnnxYoloEngine inference adapter. Each model is trained on a station-specific defect class set.

Station	Model File	Classes
2 — Stitch	models/station_2/yolo_defect.onnx	broken_thread, skip_stitch
3 — Plastic	models/station_3/plastic_crack.onnx	part_correct, part_missing, part_wrong
4 — Cosmetic	models/station_4/cosmetics.onnx	wrinkle_puckering, foam_showthrough

6.2 Classical Vision

Edge / Contour	Canny edge detection + contour analysis for dimensional measurements.
Colour Matching	BGR/HSV colour comparison against variant nominal thread colour.
NCC Template Match	Normalised Cross-Correlation golden-image match (Station 3, threshold 0.85).
Stitch Counting	Horizontal projection profile analysis to count stitches per inch.
Gap Measurement	Distance transform on binary edge mask to measure shield gap in mm.

All dimensional measurements are expressed in real-world units (mm, degrees) using the per-camera pixel_size_mm calibration factor. A **LimitsEvaluator** compares each measurement against the merged limits loaded from default_limits.yaml plus any variant-specific overrides.

Limit Types

Limit Type	Logic	Fields
RangeLimit	≤ pass_max → PASS | pass_max < v < fail_at → REVIEW | ≥ fail_at → FAIL	pass_max, fail_at
ToleranceLimit	|v – nominal| ≤ pass_tol → PASS | ≤ fail_tol → REVIEW | > fail_tol → FAIL	pass_tol, fail_tol

BooleanRule	Immediate FAIL or REVIEW on any detection; optionally required	fail_on_any, review_on_any, required

The worst-case aggregation rule applies across all defects and cameras: **FAIL > REVIEW > PASS**. A single FAIL defect anywhere in the station produces a FAIL station result regardless of other measurements.

The React/TypeScript frontend (built with Vite) is served from the same FastAPI origin at /. All components authenticate using the INSPECTION_API_KEY bearer token.

Component (file)	URL / Endpoint	Description
LiveLine.tsx	/api/v1/dashboard/live	Rolling 20-seat pass/fail strip. Shows seat ID, variant, outcome, label-printer status, and timestamp in real time.
LiveCompanion.tsx	/api/v1/dashboard/live-companion	Per-station snapshot images, defect count, cycle time, golden reference images, and variant nominal values for the latest seat.
QualityTrends.tsx	/api/v1/dashboard/quality-trends	Pass / Review / Fail rate gauges and a Pareto bar chart ranking defect types by frequency.
DefectGallery.tsx	/api/v1/dashboard/defect-gallery	Filterable gallery of all detected defects with annotated image, bounding box, confidence, measured vs nominal value.
ShiftReport.tsx	/api/v1/dashboard/shift-report	End-of-shift summary: total seats, defect breakdown, camera vs inspector comparison, false reject rate.
SystemHealth.tsx	/api/v1/dashboard/system-health	PLC connectivity, database status, operating mode (shadow vs normal), per-station online status.
ReviewQueue.tsx	/api/v1/reviews/pending	List of REVIEW-state seats. Inspector selects PASS / REWORK / SCRAP and adds notes.
AuditLog.tsx	/api/v1/audit	Paginated immutable event log filterable by seat ID.
SPCCharts.tsx	/api/v1/analytics/spc-measurements	X-bar control charts for all SPC metrics with UCL/LCL lines, Cpk readout, and violation count.
QualityAuditor.tsx	/api/v1/audit/samples	Random sampling interface for quality auditors to verify previously PASS-rated seats physically.
Settings.tsx	/api/v1/system/*	Toggle shadow mode, toggle image saving, trigger image cleanup, view storage statistics.
ROIEditor.tsx	/api/v1/stations/{id}/rois	Interactive bounding box editor to redefine Regions of Interest per camera without code changes.
StationTester.tsx	/api/v1/stations/{id}/inject-image	Upload any image to run it through a station's full AI + limits pipeline and see the annotated result.
LiveCameraStreams.tsx	/api/v1/live-inspect-proxy	Live surface analysis from a connected webcam or phone camera, showing defect overlays in near-real time.
PLCSimulator.tsx	Internal	Visual PLC simulator panel showing pallet positions, trigger states, and printer relay status.
MESIntegration.tsx	Internal	MES/ERP integration status panel for upstream production order data.

DefectTaxonomy.tsx	Internal	Reference panel showing the full defect taxonomy with example images and descriptions.

All endpoints (except /health and /) require the X-API-Key header or a Authorization: Bearer <key> header. The key must be ≥ 32 characters and is set via the INSPECTION_API_KEY environment variable. Interactive Swagger UI is available at /docs.

Endpoint	Description
POST /api/v1/stations/result	Submit a station inspection result (rate-limited 60/min).
GET /api/v1/seats/{seat_id}	Full inspection history and outcome for a seat.
GET /api/v1/seats/{seat_id}/export	ZIP bundle: JSON audit trace + defect images (rate-limited 10/min).

Endpoint	Description
GET /api/v1/reviews/pending	List inspections in REVIEW state with image paths.
POST /api/v1/reviews/{id}/decision	Submit PASS / REWORK / SCRAP decision.

Endpoint	Description
GET /api/v1/audit	Query audit log; filter by seat_id, limit up to 500.
POST /api/v1/audit/sample	Randomly sample N PASS seats for quality audit.
GET /api/v1/audit/samples	List all audit samples and their status.
POST /api/v1/audit/sample/{id}/decision	Record auditor PASS/FAIL verdict.

Endpoint	Description
POST /api/v1/shadow-mode/decision	Record an inspector's shadow decision for a seat.
GET /api/v1/dashboard/shadow-mode	Shadow mode statistics — agreement, escapes, false rejects.
GET /api/v1/dashboard/shift-report	Full end-of-shift quality report.

Live Monitoring Dashboards

Endpoint	Description
GET /api/v1/dashboard/live	Last 20 seats — pass/fail strip.
GET /api/v1/dashboard/quality-trends	Pass/review/fail rates + defect Pareto.
GET /api/v1/dashboard/defect-gallery	Defects with images; filter by station and type.
GET /api/v1/dashboard/live-companion	Latest seat snapshots + golden references.
GET /api/v1/dashboard/system-health	PLC status, operating mode, station liveness.

Analytics

Endpoint	Description
GET /api/v1/analytics/spc-measurements	SPC X-bar data with UCL/LCL/Cpk for all metrics.

Endpoint	Description
GET /api/v1/system/settings	Current runtime settings + storage stats.
POST /api/v1/system/save-images	Toggle image saving on/off (no restart needed).
POST /api/v1/system/shadow-mode	Toggle shadow mode on/off (no restart needed).
POST /api/v1/system/cleanup-images	Delete captures older than retention period (dry_run supported).

Endpoint	Description
GET /api/v1/cameras/detect	Detect connected cameras (webcam, Basler, Hikrobot).
GET /api/v1/cameras/test	Test a specific camera adapter and return metadata.
GET /api/v1/cameras/snapshot	Capture a live JPEG frame.
GET /api/v1/cameras/config	Return all station camera YAML configs.
POST /api/v1/cameras/config	Update a camera's adapter/settings in YAML (restart required).
GET /api/v1/cameras/all-status	All configured cameras and their enabled state.

Endpoint	Description
GET /api/v1/stations/{id}/rois	Fetch ROI bounding boxes for a station / camera.
POST /api/v1/stations/{id}/rois	Update ROI bounding boxes in YAML config.
POST /api/v1/stations/{id}/inject-image	Inject an image and run the full inspection pipeline.

Endpoint	Description
POST /api/v1/live-inspect-frame	Analyse a JPEG frame via Station 1 surface analyser.
GET /api/v1/live-inspect-proxy	Capture from a local camera and run Station 1 analysis.

Defect Feedback

Endpoint	Description
POST /api/v1/defects/{id}/feedback	Record inspector correction for active-learning pipeline.

Golden Images

Endpoint	Description
GET /api/v1/variants/{variant}/golden-images/{component}	Serve golden reference image.

9.1 Human Review Queue

Seats with a REVIEW outcome — where the AI is not confident enough to PASS or FAIL definitively — are placed in the human review queue. An inspector accesses the **ReviewQueue** dashboard panel, views the annotated image, and submits one of three verdicts:

PASS	Defect is within acceptable limits — pallet released to mainline.
REWORK	Defect is present but can be corrected — pallet diverted for rework.
SCRAP	Seat is non-conforming and cannot be reworked — pallet scrapped.

The decision is persisted in HumanReviewRecord and propagated to the overall seat record. The PLC is updated and the label printer is enabled/disabled accordingly. All review actions are written to the audit log.

9.2 Audit Sampling

A quality manager can request a random sample of PASS-rated seats for physical verification against a reference master. The system:

- Selects N seats from all PASS-state seats not previously sampled.
- Creates AuditSampleRecord entries with status PENDING.
- Lists them in the QualityAuditor dashboard.
- Records the auditor's PASS/FAIL verdict + discrepancy details.
- Writes all outcomes to the audit log.

9.3 Shift Reports

The shift report (GET /api/v1/dashboard/shift-report) is generated on demand. It includes:

- Total seats inspected in the current data set.
- Pass / Review / Fail counts and percentage rates.
- Camera vs inspector agreement rate (shadow mode data).
- Defect escape rate — seats where AI passed but inspector failed.
- False reject rate — seats where AI failed but inspector passed.
- Defect type breakdown — counts per defect class.
- Configurable false-reject threshold for alerting (default 3%).

Statistical Process Control is available for seven continuous measurements across the four stations. Data is retrieved from GET /api/v1/analytics/spc-measurements?metric=<key>.

Metric Key	Label	Station	Nominal	LSL	USL	Unit
stitches_per_inch	SPI	2	7.0	6.5	7.5	spi
stitch_spi_delta	Stitch Pitch Dev.	2	0.0	–0.5	+0.5	spi
seam_position_mm	Seam Position Dev.	2	0.0	–1.5	+1.5	mm
shield_gap_mm	Plastic Shield Gap	3	2.5	0.5	4.5	mm
flash_mm	Plastic Flash	3	0.0	0.0	0.5	mm
recliner_angle_deg	Recliner Angle	4	90.0	88.0	92.0	deg
track_position_mm	Track End Position	4	240	238	242	mm

Control limits are computed as mean ± 3σ. The response also returns **Cp** (process capability) and **Cpk** (process capability index) for the selected metric. If fewer than 15 real data points exist, synthetic demo data is returned so the chart is always populated.

Generated: 24 October 2026

10.1 System Configuration — configs/system.yaml

```
system:
environment: "development" # "development" or "production"
shadow_mode: false # true = shadow, false = normal
operating_mode: "shadow" # "shadow" or "camera_leads_with_audit"
save_images: true # false = no raw/annotated saves to disk
image_retention_days: 30 # captured images older than this are deleted

database:
url: "sqlite:///./data/inspection.db"
echo_sql: false

plc:
adapter: "simulator" # simulator / modbus / snap7 / opcua
host: "127.0.0.1"
port: 502
rack: 0
slot: 1
poll_interval_ms: 100

storage:
image_dir: "storage/images"
raw_image_dir: "storage/images/raw"
annotated_image_dir: "storage/images/annotated"
defect_crops_dir: "storage/images/defects"

coordinator:
expected_stations: ["STATION_1","STATION_2","STATION_3","STATION_4"]
cycle_timeout_s: 60.0
```

10.2 Seat Variants

Each variant lives in configs/variants/<VARIANT_ID>/config.yaml.

Variant ID	Description
FRONT_LH_BLACK	Front Left Hand seat — black leather/rexene (default).
FRONT_RH_BLACK	Front Right Hand seat — black.
FRONT_LH_GREY	Front Left Hand seat — grey.
REAR_BENCH_BLACK	Rear bench seat — black.

A variant config defines nominals, required components, golden images path, and limit overrides:

```
variant_id: "FRONT_LH_BLACK"
hand: LH | position: FRONT | color: BLACK

nominals:
stitch:
stitches_per_inch: 6.0
seam_position_mm: 12.0
thread_color_bgr: [210, 210, 210]
plastic:
shield_gap_mm: 2.5
mechanism:
recliner_angle_deg: 90.0
track_end_position_mm: 240.0

components:
```

```
side_shield: { expected_hand: LH, required: true }
isofix_cover: { count: 2, required: true }
clips: { count: 6, required: true }

golden_images_dir: golden_images
limit_overrides: {} # empty = default_limits.yaml applies
```

10.3 CLI Launch Options

```
python -m camerainspection.main \
--host 0.0.0.0 \
--port 8000 \
--config configs/system.yaml \
--plc-adapter simulator \
--camera-adapter replay \
--run-worker # start background pallet worker
```

10.4 Environment Variables

INSPECTION_API_KEY	Required. API key ≥ 32 chars. Must not be the default placeholder.
DATABASE_URL	Override SQLite path, e.g. postgresql://user:pw@host/db.
CORS_ALLOWED_ORIGINS	Comma-separated origin list for browser API calls.
UI_INDEX_PATH	Override the path to frontend/dist/index.html.
IMAGE_STORAGE_ROOT	Base path for image storage (default: storage/images).

11.1 Label Printer Safety Interlock

This is the primary safety gate. The label printer relay is energised if and only if *all five* conditions are simultaneously true:

#	Condition
1	All four stations reported a result for this seat.
2	All four station results are PASS (no REVIEW, no FAIL).
3	Station 4 mechanism cycle test returned PASS.
4	Station 4 mechanism includes a valid recliner_angle_deg measurement.
5	The PLC lock sensor is_mechanism_locked() returns True.

■ **The printer relay is also automatically disabled when: a new seat arrives on the line, the print-pulse timeout (5 s) elapses, a FAIL/REVIEW result is received from any station, or the E-Stop is triggered.**

11.2 Emergency Stop

Triggering the E-Stop via LineController.trigger_emergency_stop() executes these actions atomically in order:

- Stop the conveyor immediately.
- Disable the label printer (PLC output cleared).
- Assert pallet hold at ALL four stations simultaneously.
- Set operator panel to Red ON / Green OFF.
- Log a CRITICAL-level event to the application log.

Reset is via reset_emergency_stop(), which clears the flag and sets the operator panel to Amber (standby). The conveyor must be restarted manually by the operator.

11.3 Fail-Safe Error Handling

Every potential failure is handled by defaulting to the most restrictive outcome. The principle is: *if in doubt, hold the pallet*.

Failure Scenario	Automatic Response
Camera fails to connect	FAIL result, pallet held.
Image capture returns None	FAIL result, pallet held.
AI inference raises exception	FAIL result, pallet held.
Database write fails	FAIL result, pallet held, PLC notified FAIL.
Fixture clamp not confirmed	HardwareError → FAIL, pallet held.
Lighting read-back out of tolerance	HardwareError → FAIL, pallet held.
Calibration manager absent (strict mode)	FAIL defect added to result.
Missing station result at timeout	Seat auto-finalised as FAIL.
PLC communication lost	FAIL result, pallet held (no PASS reaches PLC).
Required measurement missing / NaN	station.incomplete_checks FAIL defect added.
Zero checks / measurements produced	station.no_checks_performed FAIL defect added.

A critical safety ordering is enforced: the database record is written and committed *before* the PLC is notified of a PASS result. If the DB write fails, the outcome is downgraded to FAIL and the PLC is notified FAIL, ensuring the physical line never acts on a result that isn't persisted.

11.5 API Security

API Key Validation	All write endpoints require X-API-Key (≥ 32 chars, not default). Constant-time comparison prevents timing attacks.
Rate Limiting	Write endpoints: 60 req/min per IP. Export endpoints: 10 req/min per IP.
Seat ID Validation	Regex-enforced: [A-Za-z0-9_.:-]{1,64} — prevents injection.
Path Traversal Guard	Export endpoint verifies image paths are inside repo root before adding to ZIP.
CORS	Restricted to configured origins (default: localhost ports 8000, 5500, 5173, 3000).

Feature	Description	Location
Multi-Station Pipeline	4 stations: leather, stitch, plastic, mechanism	Stations 1–4
Shadow Mode	Full pipeline without physical PLC outputs	System config / API
PatchCore AI	Anomaly detection for surface inspection	Station 1 / inference/
YOLO AI	Object detection for defects, plastics, cosmetics	Stations 2–4 / inference/
Fixture Control	Automated fixture clamp/release with confirmation	hardware/fixture/
Lighting Control	Per-camera lighting profiles with read-back	hardware/lighting/
Human Review Queue	Inspector PASS/REWORK/SCRAP interface	API + ReviewQueue.tsx
Audit Sampling	Random PASS sampling for QA auditors	API + QualityAuditor.tsx
Shift Reports	End-of-shift quality analysis with AI vs inspector	API + ShiftReport.tsx
SPC Charts	X-bar control charts with Cpk for 7 metrics	API + SPCCharts.tsx
Live Dashboard	20-seat strip, defect gallery, quality trends	Frontend dashboards
Image Export	ZIP bundle: JSON trace + annotated images per seat	GET /seats/{id}/export
ROI Editor	Interactive ROI bounding box editor	ROIEditor.tsx
Station Tester	Inject image through any station pipeline	StationTester.tsx
Active Learning	Inspector feedback stored for model retraining	API + data/training_feedback/
Variant System	Multiple seat variants with per-variant limits	configs/variants/
Immutable Audit Log	Timestamped event trail for OEM compliance	DB / GET /audit
Image Cleanup	Automatic retention-based storage management	image_cleanup.py
Camera Detection	Auto-detect webcam/Basler/Hikrobot cameras	GET /cameras/detect

Method	Endpoint	Purpose
GET	/health	Service liveness check.
GET	/	Serve SeatGuard React dashboard.
POST	/api/v1/stations/result	Ingest station result.
GET	/api/v1/seats/{id}	Full seat inspection history.
GET	/api/v1/seats/{id}/export	ZIP audit trace + images.
GET	/api/v1/reviews/pending	Review queue.
POST	/api/v1/reviews/{id}/decision	Submit review verdict.
GET	/api/v1/audit	Audit log query.
POST	/api/v1/audit/sample	Generate audit sample.
GET	/api/v1/audit/samples	List audit samples.
POST	/api/v1/audit/sample/{id}/decision	Submit audit verdict.
POST	/api/v1/shadow-mode/decision	Record shadow decision.
GET	/api/v1/dashboard/live	Live 20-seat strip.
GET	/api/v1/dashboard/quality-trends	Pass/fail rates + Pareto.
GET	/api/v1/dashboard/defect-gallery	Filterable defect gallery.
GET	/api/v1/dashboard/live-companion	Latest seat snapshots.
GET	/api/v1/dashboard/shift-report	Shift quality report.
GET	/api/v1/dashboard/shadow-mode	Shadow mode stats.
GET	/api/v1/dashboard/system-health	System health.
GET	/api/v1/analytics/spc-measurements	SPC chart data.
GET	/api/v1/system/settings	Runtime settings.
POST	/api/v1/system/save-images	Toggle image saving.
POST	/api/v1/system/shadow-mode	Toggle shadow mode.
POST	/api/v1/system/cleanup-images	Cleanup old images.
GET	/api/v1/cameras/detect	Detect cameras.
GET	/api/v1/cameras/test	Test camera.
GET	/api/v1/cameras/snapshot	Capture JPEG frame.
GET	/api/v1/cameras/config	Camera YAML configs.
POST	/api/v1/cameras/config	Update camera config.
GET	/api/v1/stations/{id}/rois	Get ROI definitions.
POST	/api/v1/stations/{id}/rois	Update ROI definitions.
POST	/api/v1/stations/{id}/inject-image	Inject test image.
POST	/api/v1/live-inspect-frame	Analyse JPEG frame.

GET	/api/v1/live-inspect-proxy	Capture + analyse live.
POST	/api/v1/defects/{id}/feedback	Record defect feedback.

Column	Type	Description
id	Integer PK	Auto-increment primary key.
seat_id	String(64)	Unique seat identifier.
variant_id	String(64)	Variant code, e.g. FRONT_LH_BLACK.
outcome	String(16)	PASS / REVIEW / FAIL.
shadow_mode	Boolean	Was this result produced in shadow mode?
label_printer_enabled	Boolean	Was label printer enabled for this seat?
mechanism_cycle_passed	Boolean	Did Station 4 mechanism cycle pass?
lock_sensor_confirmed	Boolean	Was PLC lock sensor confirmed?
created_at	DateTime	Timestamp when seat first arrived on line.
completed_at	DateTime	Timestamp when all stations reported.

Column	Type	Description
id	Integer PK	Auto-increment.
seat_id	String(64) FK	Links to SeatInspectionRecord.
station_id	String(32)	STATION_1 … STATION_4.
outcome	String(16)	PASS / REVIEW / FAIL.
cycle_time_ms	Float	Total cycle duration in milliseconds.
model_version	String(32)	AI model version used.
config_version	String(32)	Configuration version used.
raw_image_path	String	Relative path to primary raw image.
annotated_image_path	String	Relative path to annotated image.
camera_results_json	Text	JSON blob of per-camera results.
camera_image_paths_json	Text	JSON map camera_name → path.

DefectRecord

Column	Type	Description
id	Integer PK	Auto-increment.
station_result_id	Integer FK	Links to StationResultRecord.
defect_type	String(64)	Defect class name.
outcome	String(16)	PASS / REVIEW / FAIL.
measured_value	Float	Actual measured value.

nominal_value	Float	Target nominal value.
unit	String(16)	mm / deg / spi / etc.
confidence	Float	AI confidence score.
bbox_x/y/w/h	Integer	Bounding box in image pixels.
roi_name	String	ROI where defect was found.
camera_name	String	Camera that captured the defect.
description	Text	Human-readable defect description.

Column	Type	Description
id	Integer PK	Auto-increment.
seat_id	String FK	Seat under review.
station_id	String	Station that triggered the review.
inspector_id	String	Badge ID of reviewing inspector.
decision	String	PASS / REWORK / SCRAP.
notes	Text	Inspector notes.
status	String	PENDING / RESOLVED.
created_at	DateTime	When review was queued.
reviewed_at	DateTime	When inspector submitted decision.

Column	Type	Description
id	Integer PK	Auto-increment.
seat_id	String	Associated seat (may be NULL for system events).
station_id	String	Associated station (may be NULL).
event_type	String(64)	STATION_RESULT_SUBMITTED, HUMAN_REVIEW_RESOLVED, etc.
details	Text	Full event details string.
timestamp	DateTime	UTC timestamp of event.